Datastream - Michaël Bettan

Definition: Serverless Change Data Capture (CDC) and Replication Service that allows you to synchronize data across heterogeneous databases, storage systems, and applications reliably and with minimal latency.	**Availability**:	**SLA**: >=99.9%

Use Cases:
- *Analytics*
- *Database Replication (e.g., Cloud SQL → Cloud Spanner)*
- *Event-driven architectures*

***Data integration**: Data streams from databases and Software-as-a-Service (SaaS) cloud services can feed a near-real-time data integration pipeline by loading data into BigQuery via Dataflow or Cloud Data Fusion.*

***Streaming analytics**: Changes in databases are ingested into streaming pipelines such as with Dataflow for fraud detection, security event processing, and anomaly detection.*

***Near-real-time availability of data changes**: Availability of data changes in near-real-time powers artificial intelligence and machine learning applications to prevent churn or increase engagement via marketing efforts or by feeding back into production systems.*

Billing: Based on gigabytes (GBs) processed. Usage is billed in per-byte increments, on a per-stream basis, and is stated in GB (500 MB is 0.5 GB, for example). Bytes are counted, based on raw (uncompressed) data.
- CDC
- Backfill

In addition to the cost of data processed by Datastream, you're billed for any resources that you use to transfer, store, or process your data: Cloud Storage, Dataflow and Networking

Change Data Capture (CDC) definition

- *Each change is a source system that is captured and recorded in a data store. This is helpful in cases where it is important to know all changes over time and not just the state of the database at the time of data extraction.*
  - *If you need to know only the final value, then an ETL or ELT process is sufficient*
  - *If you need to know all the changes in value then a change data capture approach is appropriate*
- Alternative of Federated Queries:
  - Query external data directly from BigQuery
  - No data replication required
  - Best for **occasional access and joining data**

Key features

- **Serverless**: configure a stream and the data starts moving. There are no installation, resource allocation, or maintenance overheads. As data volumes grow and shrink, Datastream's auto scaling capabilities allocate resources to keep data moving in real time, automatically.
- **Unified Avro-based type schema**: Datastream enables simple, source-independent processing by converting all source-specific data types into a unified Datastream type schema, based on Avro types.
- **Stream historical and CDC data**: Datastream streams both historical and CDC source data in real time, simultaneously.
- **Oracle CDC without additional licenses**: Datastream provides LogMiner-based CDC streaming from any Oracle source version 11.2g and above, without the need to pay for additional licenses or software installations.
- **Cloud Storage destination**: CDC data is written to self-describing Avro or JSON files in Cloud Storage continually. This information is easily consumable for additional processing, either directly in place or by loading downstream to another destination such as BigQuery.

Key concepts

- Datastream ingests a lot of data in real time from a variety of sources, and makes the data available for consumption in the destination.
- The unit of data stored by Datastream is an event. A stream represents continuous ingestion of events from a source and writing them to a destination.
- **Private connectivity configurations** enable Datastream to communicate with data sources over a secure, private network connection. This communication happens through Virtual Private Cloud (VPC) peering.
- **Connection profiles** represent connectivity information to a specific source or destination database.
- **Streams** represent a source and destination connection profile pair, along with stream-specific settings.
- **Objects** represent a sub-portion of a stream. For instance, a database stream has a data object for every table being streamed.
- **Events** represent every data manipulation language (DML) change for a given object.

Source & Destination

- **Sources**: Oracle and MySQL
- **Destination:** Cloud Storage (change streams)
- Limitation to Cloud Storage destination. Requires Dataflow template for custom destination workarounds:
  - **Replication:** Cloud SQL, Spanner, BigQuery, Bigtable, MongoDB, Databricks

Connectivity options

- Private connectivity configurations define a peering connection between a VPC network and Google's private network, letting Datastream communicate over internal IP addresses. Private connectivity establishes a dedicated connection on the Datastream network, meaning no other customers share it
- VPC Peering
- IP ALLOW-LIST
- Port-forwarding via SSH tunnel
- Private connectivity is optional, Datastream also supports other modes of connectivity over public networks.